

3 Partial Derivatives

3.5.1 Implicit Differentiation

We can use the chain rule to obtain an application to implicit differentiation from Calculus I:

Example 3.5.1 Find y' if $x^3 + y^3 = 6xy$.

3.6 Directional Derivatives and The Gradient Vector

Definition 3.6.1 The directional derivative of $f(x, y)$ at (x_0, y_0) in the direction of a unit vector $\vec{u} = \langle a, b \rangle$, is given by

$$D_{\vec{u}}f(x_0, y_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + ha, y_0 + hb) - f(x_0, y_0)}{h}.$$

Theorem 3.6.2 *If f is a differentiable function of x and y , then f has a directional derivative in the direction of any unit vector $\vec{u} = \langle a, b \rangle$, and*

$$D_{\vec{u}}f(x, y) = f_x(x, y)a + f_y(x, y)b.$$

3.6.1 The Gradient Vector

Notice that we could write the formula for the directional derivative $D_{\vec{u}}f(x, y)$ as

$$\langle f_x(x, y), f_y(x, y) \rangle \cdot \vec{u}.$$

Definition 3.6.3 We call the vector

$$\langle f_x(x, y), f_y(x, y) \rangle$$

the gradient vector of f at (x, y) . It is denoted ∇f .

Example 3.6.4 If $f(x, y)$ is given by

$$f(x, y) = \sin(x) \cos(y),$$

find the directional derivative of f at the point $(0, 0)$ in the direction $\vec{u} = \langle 1, 1 \rangle$.

Example 3.6.5 If $f(x, y, z) = x \sin(yz)$, find the gradient of f and the directional derivative of f at $(1, 3, 0)$ in the direction of $\langle 1, 2, -1 \rangle$.

Theorem 3.6.6 Suppose that f is a differentiable function of two or three variables. The maximum value of the directional derivative is given in the direction of the gradient of f , and its value is $|\nabla f|$.

3.6.2 Tangent Planes to Level Surfaces

Definition 3.6.7 Given a surface defined by

$$F(x, y, z) = k,$$

the tangent plane to the surface at the point (x_0, y_0, z_0) is given by

$$\frac{\partial F}{\partial x}(x_0, y_0, z_0)(x - x_0) + \frac{\partial F}{\partial y}(x_0, y_0, z_0)(y - y_0) + \frac{\partial F}{\partial z}(x_0, y_0, z_0)(z - z_0) = 0,$$

Example 3.6.8 Find the tangent plane to the surface defined by

$$\frac{x^2}{4} + y^2 + \frac{z^2}{9} = 3$$

at the point $(-2, 1, -3)$.